Control Plan — Part 34521-A (Hydraulic Actuator Housing)

Rev C · effective 2025-08-01 · supersedes Rev B (2024-02-15). Customer: Vortex Motors (CUST-VTX). Issued per IATF 16949:2016 §8.5.1.1.


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machine_ids: [BSAW-03, NLX-08, NLX-09]
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1. Part and process identification

Field	Value
Part number	`34521-A`
Part name	Hydraulic actuator housing
Customer	Vortex Motors (CUST-VTX) — per CSR `QMS-CSR-VTX-001`
Material	AISI 4140 alloy steel (`MAT-4140`) — incoming spec ASTM A29
Approved supplier(s)	SUP-OHIO-STEEL (primary)
Operations covered	OP10, OP20, OP30, OP40
Manufacturing plant	Plant 1 — Cleveland, OH (Auto-Turn-1 cell)

2. Control plan matrix

Op	Process	Machine	Characteristic	Spec / tolerance	Method	Sample size	Frequency	Reaction plan

OP10	Stock saw cutoff	`BSAW-03`	Bar length	250 mm ±0.5 mm	Caliper	3	Per lot	Re-cut and re-measure; quarantine bar lot if 3 consecutive fail.
OP20	Rough turn OD	`NLX-08` / `NLX-09`	OD	Ø 50.0 ±0.1 mm	Micrometer	5	Every 25 pcs	Recalibrate offset; quarantine pieces produced since last passing check.
OP30	Finish turn + bore	`NLX-08` / `NLX-09`	**Bore diameter**	**Ø 38.000 +0.025 / −0.000 mm**	Bore gauge (Mitutoyo 511 series)	5	**Every 10 pcs**	**Stop process, quarantine all pieces back to last passing check, run containment SPC, open NCR. Reaction owner: Cell Lead. Engineering disposition within 4 h per IATF 16949 §10.2.1. See PFMEA `QMS-FMEA-34521A-RevB` failure mode FM-007.**
OP30	Finish turn + bore	same	Bore surface finish	Ra ≤ 0.8 µm	Profilometer	1	Per setup	Replace insert; verify before resuming production.
OP40	Final inspection	CMM (Zeiss Contura)	Bore Ø + concentricity to OD	Ø per OP30; concentricity 0.05 mm TIR	CMM program 34521A-CMM-001	100%	Every piece	Failing pieces routed to MRB. Concentricity failures: investigate fixture wear.

3. Bore-diameter rejection — full reaction plan

If the bore diameter measured at OP30 falls outside the **Ø 38.000 +0.025 / −0.000 mm** tolerance:

1. **Immediate (within 5 minutes):** Stop the machine. Tag the last good piece. Quarantine all pieces produced since the previous passing check (typically last 10).
2. **Containment (within 1 hour):** Cell Lead runs containment SPC on the quarantined batch with 100% bore measurement. Opens NCR in QMS referencing this control plan and PFMEA failure mode FM-007 (see `QMS-FMEA-34521A-RevB`).
3. **Root cause (within 4 hours):** Process Engineering disposition. Most common causes per PFMEA: (a) boring bar insert wear past limit; (b) thermal drift after warm-up cycle interruption; (c) work offset entry error after tooling change. Verify spindle speed, feed, and depth-of-cut against work instruction `PLM-WI-34521A-OP30-RevD`.
4. **Corrective action (within 10 working days):** Per IATF 16949:2016 §10.2.1 (see `QMS-IATF-EXCERPT-001`), formal corrective action with effectiveness verification. Update control plan and PFMEA if root cause indicates a permanent process change.
5. **Customer notification:** If any quarantined piece may have shipped, notify Vortex Motors per CSR `QMS-CSR-VTX-001` within 24 hours.

> **For correct CNC turning parameters** at OP30 (spindle speed, feed rate, depth-of-cut, tool wear limits), refer to work instruction `PLM-WI-34521A-OP30-RevD`. For OEM machine-limit envelope on the NLX-08/09, see `PDM-MAN-NLX-004`.

4. Control plan review and approval

This control plan is reviewed per IATF 16949:2016 §8.5.1.1 and §10.2. Review triggers and authority are summarised below.

Trigger	Frequency / event	Approval authority
Scheduled review	Annually, on anniversary of effective date	Quality Manager + Process Engineering Manager
Process change	On any change to tooling, machine, material, or method affecting a controlled characteristic	Quality Manager + Process Engineering Manager + Plant Manager
Corrective action	On any IATF §10.2 corrective action that updates a control method or reaction plan	Quality Manager
Customer-driven	On request from Vortex Motors per CSR §4.3	Quality Manager + Customer Quality Representative

> Rev B of this control plan (effective 2024-02-15) is **SUPERSEDED** as of 2025-08-01. Any agent or process referencing Rev B for production decisions must use Rev C. Rev B is retained for traceability only.

Document control / sign-off

Role	Name	Signature	Date

Document Owner	S. Okafor, Quality Engineering — Cleveland	(on file)	2025-08-01
Approver	S. Okafor, Quality Engineering — Cleveland	(on file)	2025-08-01